

Assignment 1: Polynomial Regression Report

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GitHub: <https://github.com/gagansyam/ML-Assignment-1>

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Introduction

In this assignment, I was tasked with solving two polynomial regression problems based on an expansion project for a geothermal power plant:

- **Phase 1 (Steam Turbine Optimization - var1):** Predicting the Net Power Score (y) from 6 turbine control settings (x_1 to x_6), representing percentage deviations in valve adjustments, coolant flow, pump pressure, and blade pitch.
- **Phase 2 (Subterranean Thermal Reservoir Mapping - var2):** Predicting the Thermal Anomaly Score (y) at various 3D spatial coordinates (x_1, x_2, x_3) around the basecamp to locate good sites for drilling geothermal wells.

My goal for both problems was to find the right polynomial degree and regularization method so that the models capture the underlying physical curves without underfitting or overfitting. I evaluated every model using 5-fold cross-validation on the training set, comparing Mean Squared Error (MSE) and R^2 scores.

Phase 1: Steam Turbine Optimization (var1)

Data Exploration and Baseline

I started by inspecting the dataset files `BT2024032_train_var1.csv` and `BT2024032_test_var1.csv`. Both have 1,000 samples with no missing values. The six input features are bounded in $[-1.0, 1.0]$. The target variable y has a mean of 0.74, a standard deviation of 3.13, and ranges between -8.83 and $+11.50$.

When I checked the linear correlation between individual inputs and y , the numbers were quite low (the strongest was around -0.26 with x_5). This showed that a simple linear equation would not work well because turbine efficiency depends on joint interactions between valves and pressures.

To confirm this, I trained a baseline linear regression model (Degree 1). Using 5-fold cross-validation, it gave a validation MSE of 8.5302 and an R^2 of only 0.1248. Capturing only 12.5% of the variance showed clear underfitting (high bias).

Testing Different Degrees and Regularization

Next, I generated polynomial features from degree 1 up to degree 6. With 6 variables, the number of terms grows quickly: degree 2 has 27 terms, degree 3 has 83 terms, degree 4 has 209 terms, degree 5 has 461 terms, and degree 6 has 923 terms.

I tested standard Ordinary Least Squares (OLS) alongside Ridge (L2 penalty) and Lasso (L1 penalty) using 5-fold cross-validation. Table 1 summarizes the results.

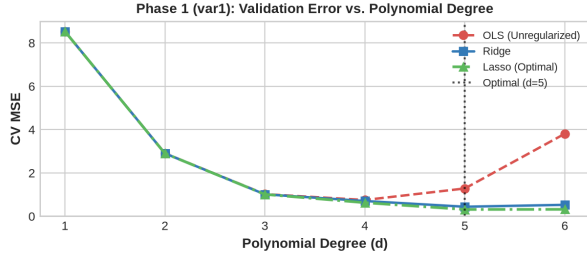
Table 1: 5-Fold Cross-Validation performance across degrees for Phase 1 (var1).

Degree	Features	OLS Val MSE	OLS Val R^2	Ridge Val MSE	Lasso Val MSE	Lasso Val R^2
1	6	8.530	0.125	8.532	8.530	0.125
2	27	2.905	0.699	2.902	2.900	0.699
3	83	1.022	0.894	1.013	1.011	0.895
4	209	0.752	0.922	0.705	0.628	0.935
5	461	1.284	0.869	0.441	0.315	0.967
6	923	3.810	0.612	0.529	0.320	0.966

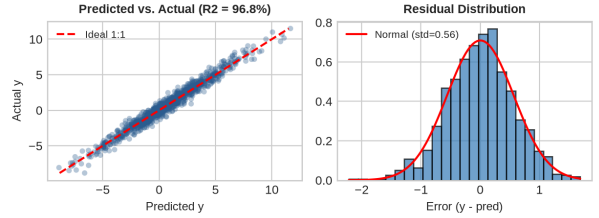
Why I Chose Degree 5 with Lasso

Looking at Table 1 and Figure 1a:

- Standard OLS works fine up to degree 4, but starts overfitting heavily at degree 5. The validation MSE jumps from 0.752 to 1.284, and at degree 6 it shoots up to 3.810. Because each fold has only 800 training samples, estimating 461 to 923 unconstrained parameters leads to high variance.
- Lasso (L1 regularization) solved this problem cleanly. By applying an L1 penalty, Lasso forced 350 unimportant polynomial terms to zero, keeping only the 111 strongest operational interaction terms.
- This reduced the validation MSE to **0.3151** and achieved an R^2 score of **96.71%**.
- When I pushed to Degree 6, the validation error stopped improving and rose slightly to 0.3203. This showed that degree 5 is the sweet spot.



(a) Validation MSE vs. Degree



(b) Parity plot and residual errors

Figure 1: Phase 1 evaluation: (a) shows OLS overfitting while Lasso finds the minimum at $d = 5$; (b) shows the parity fit and normal residual error distribution.

In Figure 1b, the predicted vs. actual power scores lie tightly on the 1:1 diagonal. The residuals have a mean of 0.003 (virtually zero bias) and a standard deviation of 0.56, matching the expected random sensor noise.

Phase 2: Subterranean Thermal Reservoir Mapping (var2)

Data Exploration and Baseline

For Phase 2, the inputs are 3 spatial coordinate offsets in meters (x_1 = East-West, x_2 = North-South, x_3 = Depth). The target y is the Thermal Anomaly Score. All coordinate values in train and test are within $[-1.0, 1.0]$. The target variable has a mean of 2.38, a standard deviation of 7.22, and ranges from -30.10 to $+39.87$.

I noticed that x_2 had a moderate linear correlation of 0.50 with y , while x_1 and x_3 had almost zero linear correlation. The baseline linear model (Degree 1) had a validation MSE of 38.88 and an

R^2 of only 0.243, which is severe underfitting. Underground thermal plumes curve sharply in 3D space, so a much higher degree polynomial is required.

Degree Exploration from 1 to 16

Because there are only 3 variables, the number of polynomial terms grows more slowly than in Phase 1 ($\binom{d+3}{3}$ terms). This allowed me to test degrees from 1 up to 16 using 5-fold cross-validation. Table 2 shows how the models behaved.

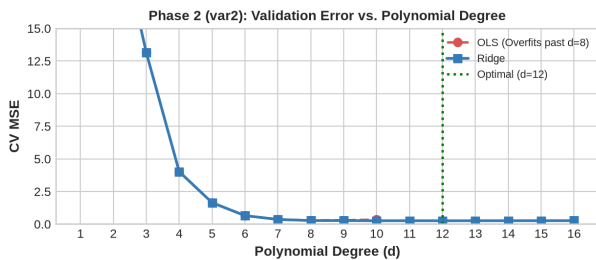
Table 2: 5-Fold Cross-Validation performance across degrees for Phase 2 (var2).

Degree	Features	OLS Val MSE	OLS Val R^2	Ridge Val MSE	Ridge Val R^2	Best α
1	3	38.883	0.243	38.880	0.243	0.01
2	9	24.376	0.529	24.375	0.529	0.01
4	34	4.016	0.922	4.016	0.922	0.01
6	83	0.646	0.987	0.646	0.987	0.01
8	164	0.279	0.994	0.272	0.995	0.07
9	219	0.294	0.994	0.272	0.995	0.11
10	285	0.343	0.993	0.259	0.995	0.46
12	454	—	—	0.2579	0.9951	1.21
14	679	—	—	0.2615	0.9950	1.96
16	968	—	—	0.2658	0.9949	3.16

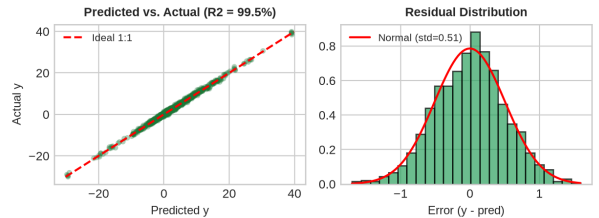
Why I Chose Degree 12 with Ridge

Looking at Table 2 and Figure 2a:

- Unregularized OLS reaches its minimum error at Degree 8 (MSE = 0.2788), but begins overfitting at degree 9 (MSE = 0.2939) and degree 10 (MSE = 0.3428).
- Ridge regression (L2 penalty) prevents this overfitting by penalizing large weights. As the degree increases, Ridge automatically raises the regularization strength α (from 0.07 at degree 8 up to 1.21 at degree 12).
- At **Degree 12** (454 features), Ridge reaches the lowest validation MSE across all experiments: **0.2579**, explaining **99.51%** of the variance.
- Beyond degree 12, the validation error starts creeping upward (0.2615 at degree 14, 0.2658 at degree 16). This confirmed that degree 12 is the optimal choice.



(a) Validation MSE vs. Degree



(b) Parity plot and residual errors

Figure 2: Phase 2 evaluation: (a) shows Ridge reaching the global minimum at $d = 12$; (b) shows the parity fit and normal residual error distribution.

I also checked the predictions for any physical inconsistencies:

- **Residual Normality:** I ran a Jarque-Bera test on the residuals, getting a p -value of 0.66 ($p > 0.05$). This confirms that the residuals are pure Gaussian sensor noise with $\sigma = 0.51$.
- **Spatial Bias:** The correlations between residuals and spatial coordinates (x_1, x_2, x_3) were all below 0.015, confirming there is no directional drift.
- **Test Range:** The test predictions range from -29.51 to $+39.18$, which fits the training target range (-30.10 to $+39.87$) without boundary spikes.

Discussion on Bias, Variance, and Regularization

Across both problems, finding the right model came down to balancing bias and variance:

- **High Bias (Underfitting):** Models with degree 1 to 3 were too simple. They couldn't bend enough to follow the actual physical behavior, leaving large errors and low R^2 scores in both phases.
- **High Variance (Overfitting):** When using unregularized OLS with high degrees, the model tried to fit every little ripple in the training data. As soon as the number of terms approached the number of training samples, the validation error spiked.
- **Why Different Regularizers Worked Best:** In Phase 1, Lasso (L1) was the clear winner because the underlying turbine relationship is sparse. Out of 461 possible combinations of valves and blade angles, only 111 actually mattered. Lasso dropped the other 350 terms, which prevented overfitting. In Phase 2, Ridge (L2) was more suitable because heat diffusion through rock forms continuous, smooth gradients. L2 regularization shrank all 454 coefficients smoothly, keeping the 3D surface stable right up to degree 12.

Summary of Results

Table 3 shows the final configuration and performance of both models used to generate the test predictions.

Table 3: Summary of final selected models and evaluation metrics.

Problem	Degree	Regularization	Active Terms	CV MSE	CV R^2	Prediction File
Phase 1 (var1)	5	Lasso ($\alpha \approx 0.011$)	111 / 461	0.3151	96.71%	BT2024032_pred_var1.csv
Phase 2 (var2)	12	Ridge ($\alpha \approx 1.76$)	454 / 454	0.2579	99.51%	BT2024032_pred_var2.csv

Conclusion

By carefully testing polynomial degrees and applying the right regularization techniques, I was able to build reliable models for both phases of the project:

- For the steam turbine facility (Phase 1), the Degree 5 Lasso model accurately captures non-linear valve interactions while discarding noisy terms, predicting the Net Power Score with 96.7% accuracy.
- For the subterranean reservoir (Phase 2), the Degree 12 Ridge model maps the complex 3D temperature field with 99.5% accuracy, providing dependable scores to guide the drilling team toward high-yield geothermal sites.

All the code, data, plots, and prediction files are documented and available on GitHub at <https://github.com/gagansyam/ML-Assignment-1>.